SOPHIE THREADGILL

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EDUCATION

New York University, New York, NY — September 2024
Bachelor of Arts, Major in Computer Science, Minor in Mathematics

Greenwich Academy, Greenwich, CT — May 2020
High Honor Roll

EXPERIENCE

Integral Ad Science, New York, NY — June 2022 – August 2022
Data Engineering Intern
- Profiled and analyzed Integral Ad Science's Twitter Brand Safety data pipeline, which gathers Tweet metadata from adjacent ad impressions to optimize digital advertisement campaign placement on the Twitter app and website
- Collaborated with a team of four interns to build a regression testing engine, which validates data pipeline components by performing both daily and ad hoc testing
- Built using Korn Shell Scripting, Ab Initio, Python, Unix Utilities, and AWS s3, and accessed Twitter API

Launch Math & Science Centers, New York, NY — July 2021 – August 2021
Instructor
- Taught elementary and middle school level summer program in chemistry and basic coding

PROJECTS

Enhancing Secret Sharing for Secure Communication (Research) — April 2024
- Conducted a research project on leakage-resilient and non-malleable secret sharing schemes to enhance security against side-channel attacks
- Optimized ratio of number of bits of leakage tolerated per share to size of the share to be an optimal leakage resilience rate of 1
- Presented findings from advanced cryptographic papers, including topics such as randomness extractors and entropy of data

Interactive Fortune Teller (Python, HTML) — April 2023
- Developed a client-server application using Python and UDP socket programming to enable real-time user interactions
- Implemented a server that processes client queries and generates randomized responses for predefined fortune-telling categories (Death, Love, Success) using Python's socket and random libraries
- Created an interactive client-side interface that accepts user inputs, communicates with the server, and displays responses

Robot Orchestra (JavaScript) — April 2022
- Developed and composed a "Robot Orchestra" performance of Beethoven's Symphony No. 5
- Used PoseNet pretrained machine learning model and ml5.js to detect motion via a neural network trained on pose data
- Processed Beethoven's symphony using Audacity by isolating audio stems and splitting them with high-pass and low-pass filters
- Synchronized musical performance across five computers, triggered by voice command and microphone input

Pixel Manipulation (Python) — February 2020
- Created a coloring book simulator deploying Canny Edge Detection, a pixel-manipulation technique used to produce a black and white version of an image where only edges are highlighted
- Produced a series of 6 images of endangered animals in an increasingly pixelated progression to parallel degradation of endangered animal population and presented to a class of students on image distortion as a method of visually representing data

Truss Modeling and Analysis (Python) — January 2020
- Collaborated with a partner to model a two-dimensional truss, and calculated largest critical force to cause structure to buckle
- Constructed a physical version of digital truss model to compare theoretical critical force to actual critical force

LEADERSHIP AND ACTIVITIES

New York Machine Learning Club, New York, NY — January 2021 – May 2024
Member, Vice President 2022

Peer Undergraduate Mentorship Program (All-Women STEM Program), New York, NY — January 2021 – May 2024
Mentee, Mentor 2023

TECHNICAL SKILLS

- Python, SQL, JavaScript, Java, HTML, C, Unix, p5.js, ml5.js, pandas/NumPy, Visual Studio Code, Jupyter Notebooks, Korn Shell Scripting, Ab Initio, PostgreSQL, LaTeX, AWS